# Introduction: thinking like an economist

Lecture notes

**What this lecture is about.** Lecture 1 had two parts. The first part explained how the course works. The second part introduced ten basic ideas that economists use to think about choices: scarcity, the cost–benefit rule, the reservation price, explicit and implicit costs, opportunity cost, the opportunity cost of money, sunk costs, marginal and average values, the rule "marginal benefit at least marginal cost", and positive and normative statements. All of them are needed for problem set 1.

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## 1 About the course

**The week.** Each problem set is due by the deadline shown on Moodle. The tutorial, with the TA Mari Edilian, is on Wednesday at 16:45. Lectures are on Wednesday and Thursday at 18:30. Each problem set is a PDF on Moodle. You may solve it in a group, but everyone enters their own answers in the Moodle quiz. The quiz is graded automatically, and the problem set is solved in the tutorial on Wednesday.

**Grading.** Problem sets count 15%, the midterm 40% and the final 45%. To pass, both conditions must hold:

$$\text{total} = 0.15 \cdot \text{PS} + 0.40 \cdot \text{midterm} + 0.45 \cdot \text{final} \;\geq\; 50\%, \qquad \frac{\text{midterm} + \text{final}}{2} \;\geq\; 50\%.$$

Problem sets cannot rescue weak exams. For example, a student with 100% on the problem sets, 50% on the midterm and 45% on the final has a total of $15 + 20 + 20.25 = 55.25\%$, but an exam average of 47.5%, and fails. A student with no problem sets, 60% on the midterm and 40% on the final has an exam average of exactly 50%, but a total of $0 + 24 + 18 = 42\%$, and also fails. The midterm is on Thursday 19.11 and the final on Thursday 28.01, both at 18:30; no aids are allowed. Moodle has the final word on dates and rooms. A harder retake with open questions follows in February; to take it, you must have taken at least one exam and handed in your problem sets.

**Materials and math.** Slides are on the course site (link also on Moodle); problem sets are on Moodle. The book, Frank, *Microeconomics and Behavior*, is optional: the lectures cover every concept needed for the problem sets and the exams. The math needed is derivatives (including partial derivatives), constrained optimization and inverse functions. The "Math recap" on Moodle (section Introduction) covers all of it. Course questions go to the Moodle forum.

**The course map.** Part 1 is the benchmark of competitive markets. Consumer theory (how people decide what to buy) gives demand; producer theory (how firms decide what to produce)

gives supply; demand and supply together give the equilibrium. This is the midterm material. Part 2 looks at markets that are not like that: one seller (monopoly), few sellers (oligopoly), side effects on others (externalities) and, if time permits, public goods. The aim is to learn the basic language of economics and simple models of the behaviour of consumers, firms and markets.

## 2 Scarcity

**Scarcity**

A resource is scarce if people want more of it than there is.

To tell whether something is scarce, imagine it were free: would there be enough for everyone? Air is free and there is enough for everyone, so air is not scarce. GPUs (the chips used to train AI models) are different: if they were free, there would be far from enough for everyone, so they are scarce.

Scarce does not mean expensive. Things have prices *because* they are scarce. And because resources are scarce, every use of a resource has an alternative that we give up. The rest of the lecture is about how to choose.

**Question 2.1: which is the scarcest?**

The AI boom needs GPUs, electricity, water for cooling and memory chips. All of them are limited: firms want more than exists. Which is the scarcest?

There is no single right answer. All four are scarce: at a price of zero, AI firms would want far more than exists. Scarcity means choosing: a GPU used here cannot be used there.

## 3 Cost–benefit

**The cost–benefit rule**

Take an action if and only if its extra benefit is at least as large as its extra cost.

Benefit and cost must be measured in the same unit, for example in euro or in hours. Examples of decisions this rule applies to: should you buy ChatGPT Plus at €20 a month? Should a researcher have 10,000 texts coded by human coders or by an LLM? Should a firm use a chatbot for customer service? Should a lab buy GPUs or rent them in the cloud?

**Question 3.1: would you subscribe?**

ChatGPT Plus costs €20 per month. You get better models, faster answers and more messages. Would you subscribe?

There is no right answer: it depends on you. "Yes" means that, for you, the benefit is at least €20 a month. "No" means that it is less than €20 a month.

## 4 Reservation price

**Reservation price**

The highest price at which you would still buy: at that price, buying and not buying are equally good.

Examples: the most you would pay per month for an AI subscription, or the most you would pay for a concert ticket. Below your reservation price, you buy. Above it, you walk away. Exactly at it, you are indifferent. So if you are indifferent between buying and not buying at a price $R$, then $R$ is your reservation price.

**Question 4.1: your reservation price?**

In early 2023, ChatGPT was free. OpenAI asked its users: at what price would it be too expensive for you? They wanted people's reservation prices, to pick the subscription price. Weeks later, ChatGPT Plus launched at $20 a month. What is your reservation price for it, in euro?

There is no right answer: it is your own number. At your reservation price you are indifferent between buying and not buying. Everyone whose reservation price is €20 or more should have answered "yes" in question 3.1.

If we sort everyone's reservation prices from highest to lowest, we get a picture that is a demand curve. Lecture 2 starts from there.

## 5 Explicit and implicit cost

**Explicit cost, implicit cost, economic cost**

An **explicit cost** is money that leaves your pocket: a payment, a bill. An **implicit cost** arises when no money changes hands, but you give up something valuable: your time, your effort, or what your own resources could earn. The **economic cost** is the sum: explicit cost + implicit cost.

The **accounting cost** counts only the explicit costs. Correspondingly there are two kinds of profit:

$$\begin{aligned}\text{accounting profit} &= \text{revenue} - \text{explicit costs},\\ \text{economic profit} &= \text{revenue} - \text{explicit costs} - \text{implicit costs}.\end{aligned}$$

**Example: Mia's start-up.** Mia quits her €70,000 job at Google to run her own LLM start-up for a year. The start-up earns a revenue of €80,000. She pays €30,000 for cloud GPUs and €12,000 for an office: explicit costs of €42,000. She gives up her Google salary of €70,000: an implicit cost. Her economic cost is $42{,}000 + 70{,}000 = €112{,}000$.

$$\begin{aligned}\text{accounting profit} &= 80{,}000 - 42{,}000 = €38{,}000,\\ \text{economic profit} &= 80{,}000 - 112{,}000 = -€32{,}000.\end{aligned}$$

The accountant sees a profit. The economist sees that Mia would be €32,000 better off staying at Google.

**Question 5.1: accounting cost and economic cost**

This weekend you visit a friend in Prague. You pay €40 for the train ticket and €60 for food and drinks. You give up your Saturday café shift, which pays €100. You neither like nor dislike the job itself.

(a) What is the accounting cost of the trip?

(b) What is the economic cost of the trip?

**Step 1: explicit costs.** The money you pay is $€40 + €60 = €100$. The accounting cost counts only these.

**Step 2: implicit costs.** You give up the €100 you would have earned at the café. No money leaves your pocket, but you lose it. Because you neither like nor dislike the job, the €100 wage is the whole value of what you give up.

**Step 3: economic cost.** Explicit plus implicit: $€100 + €100 = €200$.

(a) The accounting cost is €100. (b) The economic cost is €200.

## 6 Opportunity cost

**Opportunity cost**

The value of the best alternative you give up when you choose.

Every use of a scarce resource has an opportunity cost: its next-best use. Examples:

- TSMC's chip factory can make Nvidia AI chips or iPhone chips.
- OpenAI's GPUs can train the next model or answer your ChatGPT questions.
- Your €20 can pay for ChatGPT Plus or for dinner with friends.
- Your evening can go to the problem set or to a side job.
- A city plot can hold a data centre, housing or green space.

**Question 6.1: use the code or sell it?**

You won a one-year ChatGPT Plus gift code in a raffle. It normally costs €240. A classmate offers you €150 for the code. To you, a year of Plus is worth about €100, because you would use it only a little. Should you use the code or sell it?

**Step 1: the two options.** If you use the code, you get its value to you: €100. If you sell it, you get €150.

**Step 2: the opportunity cost of using it.** The code was free, but using it means giving up the €150 your classmate would pay. That €150 is the opportunity cost of using the code. It is not €0 ("it was free"), and it is not €240 (the list price, which nobody offers you).

**Step 3: compare.** The benefit of using the code (€100) is less than its opportunity cost (€150). Selling leaves you $150 - 100 = €50$ better off.

Sell the code. Its opportunity cost is €150, and it is worth only €100 to you.

## 7 Opportunity cost of money

**Opportunity cost of money**

Money used one way gives up the interest it could earn in a savings account.

**Example.** You have €1,000 in savings. A savings account pays 3% a year, that is $0.03 \times 1{,}000 = €30$. You keep the money in your current account at 0% instead. This costs you €30 a year, although nobody charged you anything. You do not pay the €30; you just do not get it. It is an implicit cost: no bill, but still a cost. The same holds when you lend money to a friend for free, pay a year in advance, or keep cash at home.

**Question 7.1: what does the loan cost you?**

Your friend wants to borrow €1,000 for a new laptop, for one year, interest-free. You are sure they will pay you back. Your savings account pays 5% a year. What does the loan cost you?

**Step 1.** You get the €1,000 back in full, so the only thing you lose is the interest the money would have earned in your savings account during that year.

**Step 2.** That interest is $0.05 \times €1{,}000 = €50$.

The loan costs you €50: the interest you give up for one year.

## 8 Sunk cost

**Sunk cost**

Money already spent that you can't get back: it's gone either way, so ignore it when you decide.

Examples: a concert ticket that you cannot refund or resell, or a non-refundable six-month membership for Vienna's swimming pools. Opportunity costs are about the decisions you face now. The sunk payment cannot be changed by any decision, so it does not enter. Going to the concert still costs you your evening. If you could resell the ticket, going would also cost you the resale price.

**Question 8.1: how much is sunk?**

Last month you bought an iPhone for €1,000. Today a new model comes out. You could resell your iPhone for €700 on a second-hand site. How much of the €1,000 is sunk?

**Step 1.** Only what you cannot get back is sunk. Of the €1,000 you paid, you can still get €700 back by reselling.

**Step 2.** The sunk part is $€1{,}000 - €700 = €300$.

**Step 3: what the €700 means.** The €700 resale value is still yours. Keeping your iPhone costs you that €700, because you could have the money instead.

€300 of the €1,000 is sunk.

**A follow-up.** Suppose the new model costs €1,100. Switching means selling your phone for €700 and paying €1,100, so it costs $1{,}100 - 700 = €400$. Switch only if the new phone is worth more than €400 extra to you. The thought "if only I had waited" does not matter: the €300 is gone either way.

# 9 Marginal and average

**Marginal and average**

The **marginal** value is what one more unit adds to the total. The **average** value is the total divided by the number of units.

**Example.** Spotify Family costs €18 a month for up to 6 people. With 5 people, the average cost is $18/5 = €3.60$ each. A 6th person costs €0 extra: the marginal cost is zero. Another example is one more hour of study. Decisions are about the next unit, so they need the marginal value, not the average.

**Question 9.1: how many baristas should work?**

You run a small café on campus with one coffee machine. Each barista costs you €100 per day. You tracked daily sales on days with 1, 2, 3 and 4 baristas working. More baristas serve more people, but they share one machine, so each extra barista adds less.

| baristas | sales per day |
|---|---|
| 1 | €300 |
| 2 | €520 |
| 3 | €660 |
| 4 | €720 |

What are the extra sales from the 4th barista? How many baristas should work?

**Step 1: average sales.** Average = sales ÷ number of baristas.

**Step 2: marginal sales.** Marginal = sales now − sales with one barista fewer.

| baristas | sales per day | average | marginal |
|---|---|---|---|
| 1 | €300 | $300 \div 1 = €300$ | $300 - 0 = €300$ |
| 2 | €520 | $520 \div 2 = €260$ | $520 - 300 = €220$ |
| 3 | €660 | $660 \div 3 = €220$ | $660 - 520 = €140$ |
| 4 | €720 | $720 \div 4 = €180$ | $720 - 660 = €60$ |

**Step 3: decide barista by barista.** Each barista costs €100. Barista 1 adds €300, barista 2 adds €220 and barista 3 adds €140: each adds more than €100. Barista 4 adds only €60, which is less than the €100 cost.

The 4th barista adds €60 in sales. Since $€60 < €100$, do not hire the 4th: 3 baristas should work.

The average with 4 baristas (€180 per barista) is above €100 and looks fine. But the decision is about the next barista, not the average one.

## 10 Marginal benefit $\geq$ marginal cost

**Marginal benefit $\geq$ marginal cost**

Keep increasing an activity while the next unit adds at least as much benefit as it costs.

Examples: one more slice of pizza (each tastes a little less good), one more Netflix episode tonight (it costs you sleep), one more set at the gym (each one hurts more). The extra benefit usually falls with each unit, while the cost per unit stays the same. Stop where they cross.

**Question 10.1: how many hours do you buy?**

Before your exam, an AI tutor app sells premium tutoring for €2 per hour. Assume that you can put a euro value on each hour (in real life this is hard to know; it is a simplification). The 1st hour would be worth €10 to you. Each further hour is worth €1.50 less, because you get tired and already know the basics. How many hours do you buy?

**Step 1: the value of each hour.** Start at €10 and subtract €1.50 for each further hour. Hour $n$ is worth $10 - 1.5\,(n-1)$ euro.

| hour | 1 | 2 | 3 | 4 | 5 | 6 | 7 |
|---|---|---|---|---|---|---|---|
| worth (€) | 10 | 8.50 | 7 | 5.50 | 4 | 2.50 | 1 |
| buy at €2? | yes | yes | yes | yes | yes | yes | no |

**Step 2: compare with the price.** Buy another hour while it is worth at least its €2 price. Hour 6 is worth €2.50 $\geq$ €2. Hour 7 is worth only €1 $<$ €2.

Buy 6 hours.

## 11 Positive and normative

**Positive and normative statements**

A positive statement says what is and can be checked with facts; a normative statement says what should be.

A positive statement can be checked with data. A normative statement is a value judgement: it contains words such as “should”, “too high” or “fair”. A positive statement can be false and still be positive, because it can be checked. Economists try to keep the two apart: first what is, then what should be.

**Question 11.1: positive or normative?**

Classify each statement as positive or normative.

(a) AI data centres raise electricity prices.

(b) Governments should limit AI's water use.

(c) Chip export bans slow China's AI progress.

(d) AI firms should pay artists whose work trains their models.

(e) LLMs label texts as accurately as human coders.

(f) Universities should ban ChatGPT in exams.

(a) positive, (b) normative, (c) positive, (d) normative, (e) positive, (f) normative. Statements (a), (c) and (e) describe what is, and data could show whether they are true. Statements (b), (d) and (f) say what should be.

## 12 Summary

- **Scarcity:** people want more than is available, so we must choose.
- **Cost–benefit:** take an action only if its extra benefit is at least its extra cost.
- **Reservation price:** the most you would pay; at that price you are indifferent.
- **Explicit and implicit cost:** money you pay, plus what you give up without paying.
- **Opportunity cost:** the value of the best alternative you give up.
- **Opportunity cost of money:** idle or lent money loses the interest it could earn.
- **Sunk cost:** gone whatever you do, so ignore it when deciding.
- **Marginal and average:** decide with the next unit, not the average.
- **Marginal benefit $\geq$ marginal cost:** keep going while the next unit is worth its cost.
- **Positive and normative:** what is (can be checked) versus what should be.